

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**  
**ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION**

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|-------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------|
| <b>Course Code:</b>     | ITA03                                                                                                             | <b>Course Title:</b> | Mobile computing                           |
| <b>CO Assessed:</b>     | CO3 – Precision (BL3)                                                                                             | <b>Assessment:</b>   | Assessment Tool 2 – Technical Presentation |
| <b>Weightage:</b>       | 30 % of CIA                                                                                                       | <b>Total Marks:</b>  | 20 (Answer ANY ONE)                        |
| <b>CO3 Statement:</b>   | <i>Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)</i> |                      |                                            |
| <b>Student Name:</b>    | G Mythili                                                                                                         | <b>Reg. No.:</b>     | 192372250                                  |
| <b>Section / Batch:</b> | SLOT D                                                                                                            | <b>Date:</b>         | 06-08-2026                                 |

**Code of Conduct**

*I \_\_\_\_\_ G Mythili, 192372250\_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: \_\_\_\_\_mythili\_\_\_\_\_

**Instructions**

- Duration: 10 minutes

| <b>Faculty In-charge</b> | <b>Course Coordinator</b> | <b>Head of Department</b> |
|--------------------------|---------------------------|---------------------------|
|                          |                           |                           |
| Signature: _____         | Signature: _____          | Signature: _____          |
| Date: _____              | Date: _____               | Date: _____               |

## Questions

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1. Derive the total control-channel bandwidth for a system with 150 control channels at 10 kHz each, and then calculate what percentage this represents of a 25 MHz total spectrum allocation. Show both calculations with clear steps. Discuss why this percentage matters to an operator — i.e., what this "signaling tax" costs them in terms of foregone revenue-generating capacity — and how it might be reduced through more efficient control-channel design.

